

# Solve this Problems in dart programing language using only Loops & Conditions

## 1. The Factorial Calculator

**Goal:** Calculate the factorial of a number. A factorial (denoted as  $n!$ ) is the product of all positive integers less than or equal to  $n$ . (e.g.,  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ).

**The Problem:**

Given an integer  $n = 6$ . Write a loop that calculates the multiplication of all numbers from 1 up to 6 and prints the final result.

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## 2. The Digit Counter

**Goal:** Count how many digits are in a specific number without converting it to a String.

**The Problem:**

Given an integer number = 98765. Count how many digits are in it (e.g. 6589423158965 = 13).

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## 3. The Collatz Conjecture

**Goal:** Implement the famous mathematical sequence ([Collatz conjecture](#)) that always ends in 1, regardless of the starting number.

**The Problem:**

Start with a number  $n = 12$ .

1. If  $n$  is even, divide it by 2.
2. If  $n$  is odd, multiply it by 3 and add 1.
3. Repeat this process until  $n$  becomes 1.
4. Print the value of  $n$  at every step.